

PROBLEM SET 2

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1. Maximum likelihood estimator (MLE)

This document analyzes a nonlinear monetary model derived from the so-called Laffer curve for the fiscal deficit. Consider the following specification for the fiscal deficit:

$$d = \theta \exp(-\alpha(\theta - n)), \quad (1)$$

where d denotes the fiscal deficit as a percentage of output, θ is the money growth rate, α represents the elasticity of money demand with respect to the opportunity cost of holding money, and n is the real output growth rate.

Given a target level of deficit $\bar{d}$ and fixed parameter values α and n , this framework leads to the nonlinear equation

$$f(\theta) = \bar{d} - \theta \exp(-\alpha(\theta - n)). \quad (2)$$

The objective of the analysis is to determine the values of the money growth rate θ that satisfy $f(\theta) \approx 0$, that is, those consistent with the desired fiscal deficit. Because the equation is nonlinear, analytical solutions are generally not available, and numerical methods must be employed to characterize the equilibrium money growth rates and examine their properties. A brief discussion of the economic foundations of this model can be found in Leyva (2024).

(1) MLE of α .

The parameter α is estimated via maximum likelihood using the linear regression model

$$\ln(h_t) = \text{constant} - \alpha r_t + \varepsilon_t, \quad (3)$$

where $h_t = \frac{M_t}{P_t Y_t}$ denotes the ratio of money balances to nominal GDP, r_t is the nominal interest rate on CETES-28 days, and ε_t is an error term. The estimation is carried out using the dataset `laffer.xlsx`.

The previous model can be expressed as a simple linear regression

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad (4)$$

where $y_i = \ln(h_t)$, $\beta_1 = -\alpha$, $x_i = r_t$, β_0 is the constant and ε_i the error term.

Assuming that the terms ε_i are i.i.d. $N(0, \sigma^2)$, the log-likelihood function of $\boldsymbol{\gamma} = (\beta_0, \beta_1, \sigma^2)^T$ is given by

$$\ell(\boldsymbol{\gamma} ; \mathbf{y}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta}), \quad (5)$$

To estimate $\boldsymbol{\beta} = (\beta_0, \beta_1)^T$ via maximum likelihood we need to maximize (5). A natural approach is to set its first derivative equal to zero, that is, to determine the values of $\boldsymbol{\gamma}$ such that

$$\nabla \ell(\boldsymbol{\gamma}) = \mathbf{0}. \quad (6)$$

Since the first-order conditions generally do not admit a closed-form solution when written in vector form, a convenient approach is to use Newton's method to obtain a numerical approximation of $\boldsymbol{\gamma}$.

Let

$$F(\boldsymbol{\gamma}) = \nabla \ell(\boldsymbol{\gamma}).$$

We approximate $F(\boldsymbol{\gamma})$ by a first-order Taylor expansion around $\boldsymbol{\gamma}_k$:

$$\nabla \ell(\boldsymbol{\gamma}) \approx \nabla \ell(\boldsymbol{\gamma}_k) + \nabla^2 \ell(\boldsymbol{\gamma}_k)(\boldsymbol{\gamma} - \boldsymbol{\gamma}_k), \quad (7)$$

where $\nabla^2 \ell(\boldsymbol{\gamma}_k)$ denotes the Hessian matrix of second derivatives evaluated at $\boldsymbol{\gamma}_k$.

Setting the gradient equal to zero yields the following equation:

$$\nabla^2 \ell(\boldsymbol{\gamma}_k)(\boldsymbol{\gamma} - \boldsymbol{\gamma}_k) \approx -\nabla \ell(\boldsymbol{\gamma}_k). \quad (8)$$

This expression motivates an iterative procedure. In particular, the Newton step is obtained by solving

$$\nabla^2 \ell(\boldsymbol{\gamma}_k) \mathbf{s}_k \approx -\nabla \ell(\boldsymbol{\gamma}_k). \quad (9)$$

where $\mathbf{s}_k = \boldsymbol{\gamma}_{k+1} - \boldsymbol{\gamma}_k$.

From (5) we can obtain $\nabla\ell(\boldsymbol{\gamma})$ and $\nabla^2\ell(\boldsymbol{\gamma})$ analytically:

$$\nabla\ell(\boldsymbol{\gamma}) = \begin{bmatrix} \nabla_{\boldsymbol{\beta}}\ell(\boldsymbol{\gamma}) \\ \frac{\partial\ell(\boldsymbol{\gamma})}{\partial\sigma^2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sigma^2}X'(\mathbf{y} - X\boldsymbol{\beta}) \\ -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta}) \end{bmatrix}. \quad (10)$$

$$\nabla^2\ell(\boldsymbol{\gamma}) = \begin{bmatrix} \nabla_{\boldsymbol{\beta}\boldsymbol{\beta}}^2\ell(\boldsymbol{\gamma}) & \nabla_{\boldsymbol{\beta}\sigma^2}^2\ell(\boldsymbol{\gamma}) \\ \nabla_{\sigma^2\boldsymbol{\beta}}^2\ell(\boldsymbol{\gamma}) & \frac{\partial^2\ell(\boldsymbol{\gamma})}{\partial(\sigma^2)^2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sigma^2}X'X & -\frac{1}{\sigma^4}X'(\mathbf{y} - X\boldsymbol{\beta}) \\ -\frac{1}{\sigma^4}(\mathbf{y} - X\boldsymbol{\beta})'X & \frac{n}{2\sigma^4} - \frac{1}{\sigma^6}(\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta}) \end{bmatrix}. \quad (11)$$

Under these specifications, MATLAB was used to obtain a numerical approximation of the parameter vector $\boldsymbol{\gamma}$. The Newton algorithm was implemented with a tolerance level $\text{tol} = 10^{-8}$, such that the procedure stops whenever $\|\mathbf{s}_k\| < \text{tol}$, or when a fixed maximum number of iterations is exceeded, namely $k > k_{\max} = 50$.

As an initial value for the algorithm, we set

$$\boldsymbol{\gamma}_0 = (\bar{y}, 0, S_y)$$

where $\bar{y}$ denotes the sample mean of y and S_y its sample variance.

The numerical optimization procedure yields the following maximum likelihood estimates:

$$\hat{\boldsymbol{\gamma}}_{ML} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\sigma}^2 \end{bmatrix} = \begin{bmatrix} 2.7098 \\ -2.9614 \\ 0.0989 \end{bmatrix}.$$

Thus, the estimation of α by maximum likelihood is

$$\hat{\alpha}_{ML} = 2.9614$$

(2) Showing that σ_{ML}^2 is a biased estimator.

Recall the log-likelihood function:

$$\ell(\gamma; \mathbf{y}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\mathbf{y} - X\boldsymbol{\beta})'(\mathbf{y} - X\boldsymbol{\beta}), \quad (12)$$

By differentiating this function with respect to σ^2 , we obtain:

$$\frac{\partial \ell(\gamma)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Setting this equation equal to zero, and considering the optimum values $\hat{\beta}_0$ and $\hat{\beta}_1$, it follows that the maximum likelihood estimator is given by

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \frac{1}{n} SS_{Res}$$

We know that the OLS estimator of σ^2 is unbiased. For a simple regression model, where we have only two betas, this OLS estimator is given by

$$\hat{\sigma}_{OLS}^2 = \frac{1}{n-2} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \frac{1}{n-2} SS_{Res}$$

and

$$\mathbb{E}[\hat{\sigma}_{OLS}^2] = \sigma^2.$$

Hence, we know that

$$\mathbb{E}[SS_{Res}] = (n-2)\sigma^2.$$

From which it can be obtained that

$$\mathbb{E}[\hat{\sigma}_{ML}^2] = \frac{1}{n} \mathbb{E}[SS_{Res}] = \frac{n-2}{n} \sigma^2$$

Thus, the maximum likelihood estimator is biased.

For the model presented above, the following estimates of σ^2 were obtained in MATLAB:

$$\hat{\sigma}_{ML}^2 = 0.0989, \quad \hat{\sigma}_{OLS}^2 = 0.1004.$$

2. Presentation

(1) OLS estimator of α .

We already established that the model given in (3) can be expressed as a simple regression:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad (13)$$

where $y_i = \ln(h_t)$, $\beta_1 = -\alpha$, $x_i = r_t$, β_0 is the constant and ε_i the error term.

The OLS estimator of β_1 is given by

$$\hat{\beta}_1^{\text{OLS}} = \frac{S_{xy}}{S_{xx}}$$

where

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$
$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

Using MATLAB we obtain that the estimation is $\hat{\beta}_1^{\text{OLS}} = -2.9614$, so the estimation for α is

$$\hat{\alpha}_{\text{OLS}} = 2.9614$$

As expected, the estimation of α by OLS is the same as its estimation by maximum likelihood.

(2) Plot of the demand function.

With the OLS estimation of β_1 , the estimation of β_0 can be obtained as follows:

$$\hat{\beta}_0^{\text{OLS}} = \bar{y} - \hat{\beta}_1^{\text{OLS}} \bar{x}.$$

Thus, with these values, the following regression line was obtained:

$$\hat{y} = 2.7098 - 2.9614x.$$

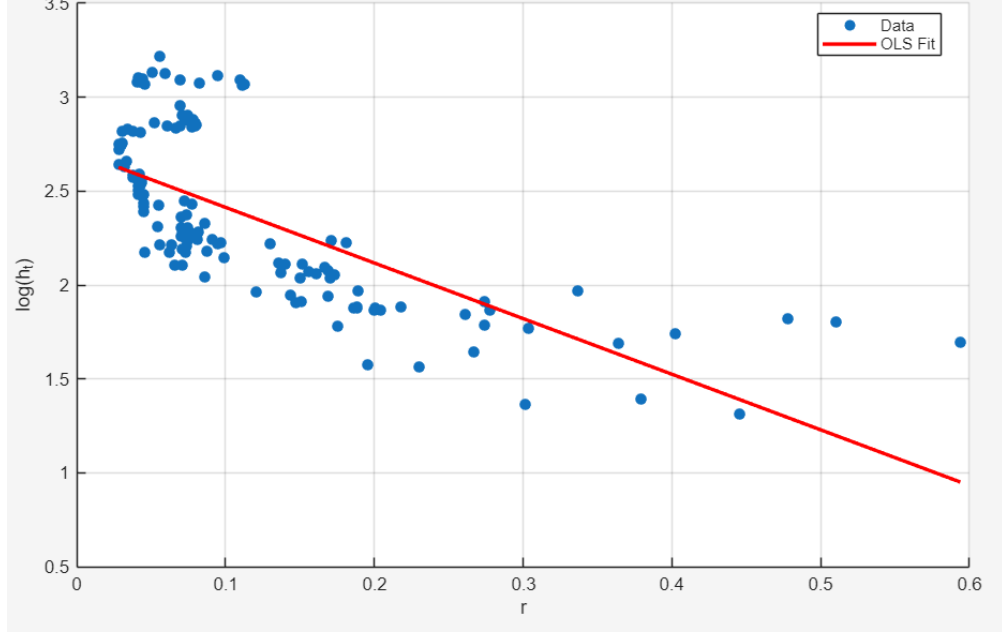


Figure 1: Demand function: OLS fit

(3) Residual-based bootstrap.

Using the OLS estimates of β_0 and β_1 , the residuals can be obtained as

$$e_i = y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i), \quad i = 1, \dots, n.$$

With these n residual values, bootstrap resampling can be applied to select a new sample of size n :

$$e_1^*, e_2^*, \dots, e_n^*.$$

Using this new bootstrap sample of residuals, together with the original x_i values, new response values can be constructed as

$$y_i^* = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i^*.$$

For this new bootstrap sample, a new estimate $\hat{\beta}_1^*$ is obtained. Repeating this procedure B times yields a bootstrap sample of β_1 , that is, a sample of α :

$$\hat{\alpha}^{*(1)}, \hat{\alpha}^{*(2)}, \dots, \hat{\alpha}^{*(B)}.$$

Using $B = 100$, the following sampling distribution of $\hat{\alpha}$ is obtained.

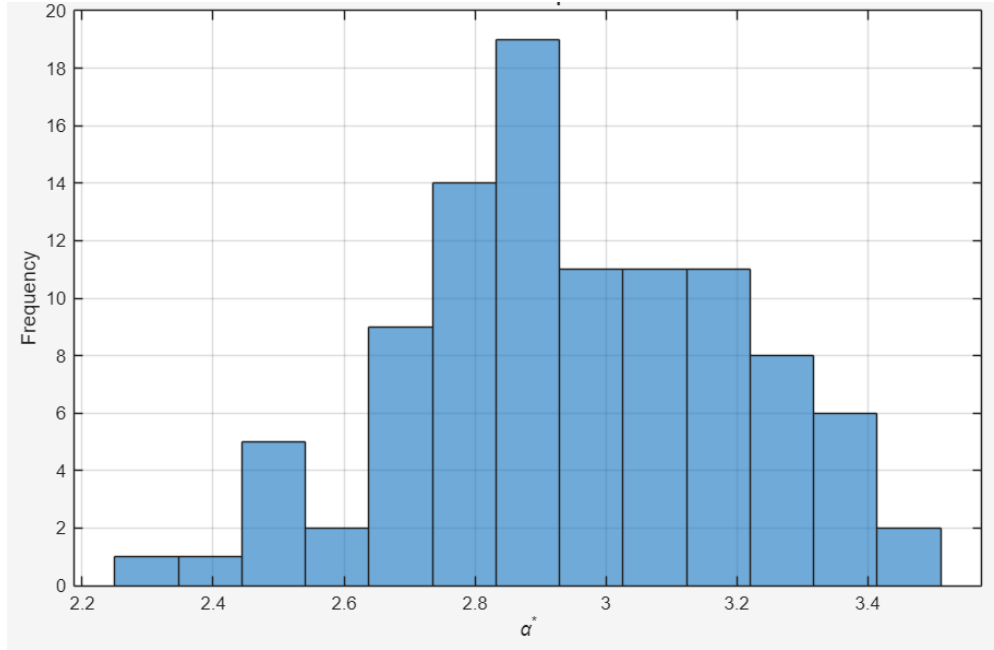


Figure 2: Residual-based bootstrap distribution of α

The mean of this distribution was also obtained: 2.9501, which is close to the OLS estimation of α .

(4) Paired bootstrap.

Note that in the previous procedure the residuals e_i were resampled in order to obtain new observations y_i^* . Another method consists of applying bootstrap resampling directly to the data pairs (x_i, y_i) . In this case, a bootstrap sample is obtained as

$$(x_1^*, y_1^*), \dots, (x_n^*, y_n^*).$$

Using this new bootstrap sample, the regression parameters are estimated again. Repeating this procedure B times yields a bootstrap sample of $\hat{\beta}_1$, i.e: a sample of $\hat{\alpha}$, which again can be expressed as

$$\hat{\alpha}^{*(1)}, \hat{\alpha}^{*(2)}, \dots, \hat{\alpha}^{*(B)}.$$

Using $B = 100$, we obtained the sampling distribution of $\hat{\alpha}$:

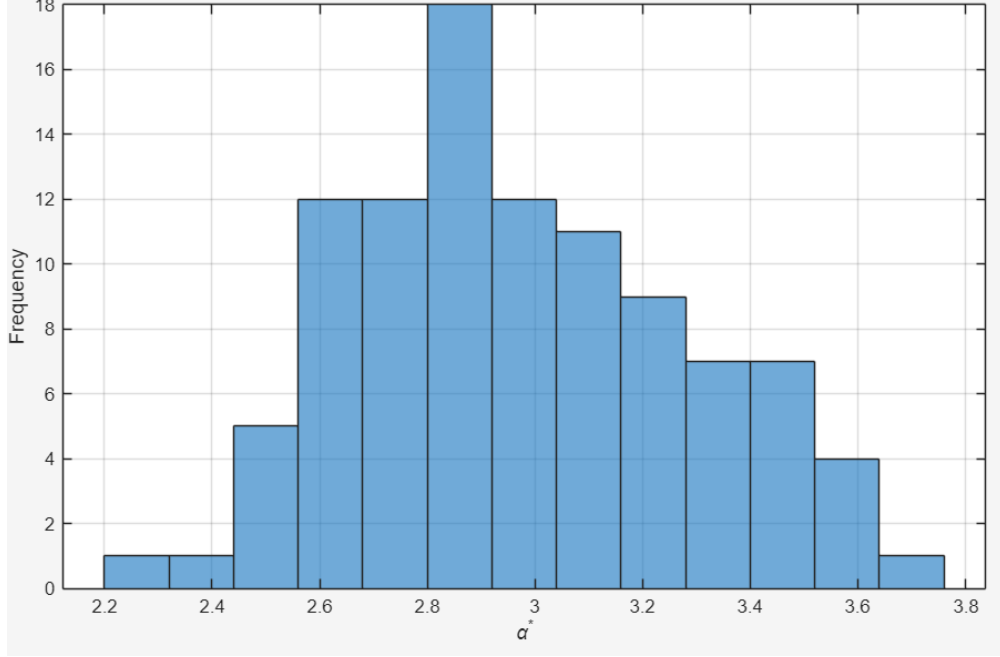


Figure 3: Paired bootstrap distribution of α

The mean of this distribution is 2.9805.

(5) Efron confidence intervals.

For each bootstrap method, the Efron (percentile) confidence interval for the parameter of interest $\alpha = -\beta_1$ is obtained from the empirical quantiles of the bootstrap replicates

$$\hat{\alpha}^{*(1)}, \hat{\alpha}^{*(2)}, \dots, \hat{\alpha}^{*(B)}.$$

Let $\hat{\alpha}_{(1)}^* \leq \hat{\alpha}_{(2)}^* \leq \dots \leq \hat{\alpha}_{(B)}^*$ denote the sorted bootstrap sample. Then the $100(1 - \gamma)\%$ Efron percentile interval is

$$\left[\hat{\alpha}_{(\lceil B\gamma/2 \rceil)}^*, \hat{\alpha}_{(\lceil B(1-\gamma/2) \rceil)}^* \right].$$

In particular, for a 95% interval:

$$CI_{0.95}^{\text{Efron}} = \left[\hat{\alpha}_{(\lceil 0.025B \rceil)}^*, \hat{\alpha}_{(\lceil 0.975B \rceil)}^* \right].$$

When $B = 100$, these correspond approximately to the 2.5th and 97.5th percentiles, i.e. the lower endpoint is around the 3rd order statistic and the upper endpoint around the 98th order statistic (depending on the quantile convention used).

We obtained the Efron confidence intervals:

$$95\% \text{ CI (Residual Bootstrap)} : [2.5043, 3.3960]$$

$$95\% \text{ CI (Paired Bootstrap)} : [2.4733, 3.5585]$$

The paired bootstrap interval is larger than the residual one (1.0852 vs. 0.8917). Nevertheless, we need to acknowledge that using $B=100$ is relatively small. With such a small number of replications, the estimated endpoints of a 95% percentile interval can have noticeable variability, so the reported confidence intervals may change if the bootstrap is rerun with a different random seed

The residual bootstrap resamples the fitted residuals $\{e_i\}$ and constructs $y_i^* = \hat{\beta}_0 + \hat{\beta}_1 x_i + e_i^*$, keeping the original x_i fixed. This method is most appropriate when the regression model is correctly specified and the errors are i.i.d. (in particular, approximately homoscedastic). In contrast, the paired bootstrap resamples the observed pairs (x_i, y_i) to form $(x_1^*, y_1^*), \dots, (x_n^*, y_n^*)$ and then refits the regression. This method can be more robust when heteroscedasticity is present, although it may be less efficient when x_i is a fixed design.

To quickly verify if heteroscedasticity is presented in this model, we can plot the regressor x vs. the squared residuals e^2 :

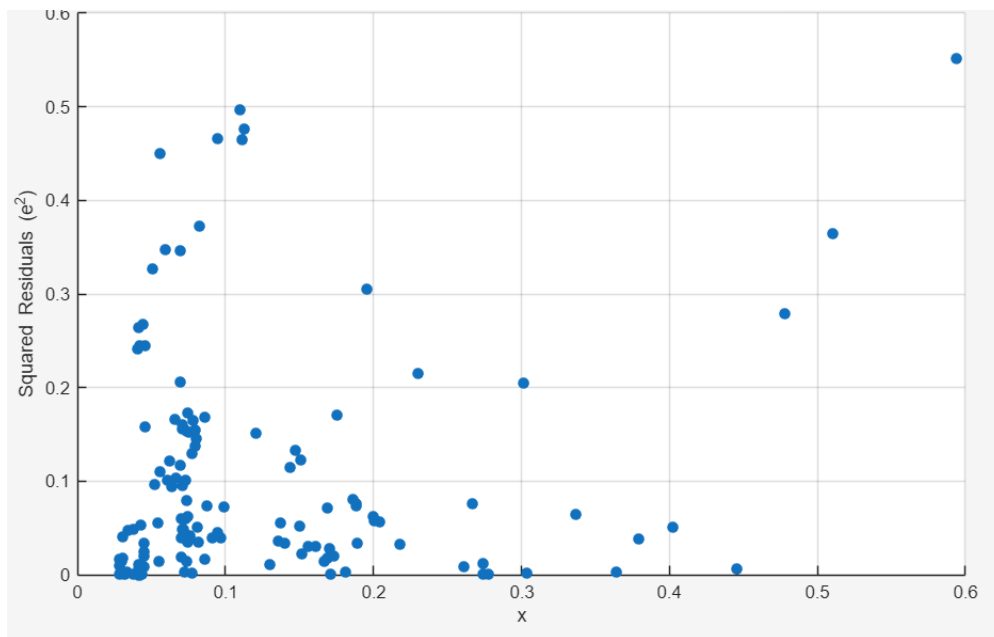


Figure 4: Regressor vs. squared residuals

It can be observed that the scatter plot does not display a homogeneous cloud of points; therefore, it cannot be asserted that the model satisfies the homoscedasticity assumption. For this reason, it may be preferable to rely on the paired bootstrap estimation.

References

Leyva, G. (2024): “Bruno and Fischer (1990): Mathematical Derivation,” Handout for Theory and Monetary Policy, ITAM, Fall 2024.